Transient Heat Conduction in an Infinite Solid Cylinder with Convection

Question

We shall plot $\theta(\xi, \tau)$ for $0 \leq \xi \leq 1, 0 \leq \tau \leq 1.5$, and Bi = 0.5 using the lowest fifteen positive roots of ζ_n.

Solution

1. Governing Root Equation

$$f(x) = xJ_1(x) - B_i J_0(x) = 0$$

with $B_i = 0.5$.

Here, J_0 and J_1 are Bessel functions of the first kind.

2. First Few Roots (manual approximation)

Known roots (for Bi = 0.5) are close to the zeros of $J_0(x)$.

Using Bessel tables:

- $J_0(x) = 0$ at 2.4048, 5.5201, 8.6537, 11.7915, …
- With Bi = 0.5, roots shift slightly **below** these values.

So, approximate:

$$r_1 \approx 2.16, r_2 \approx 5.36, r_3 \approx 8.54, r_4 \approx 11.73$$

3. Coefficient c_n

$$c_n = \frac{2J_1(r_n)}{r_n(J_0(r_n)^2 + J_1(r_n)^2)}$$

Example for $r_1 \approx 2.16$.

- From Bessel tables:

$$J_0(2.16) \approx -0.14, J_1(2.16) \approx 0.58$$

- Denominator

$$(J_0^2 + J_1^2) \approx (0.14^2 + 0.58^2) = 0.358$$

Multiply by $r_1 \approx 2.16$: → 0.773

- Numerator

$$2J_1(r_1) \approx 1.16$$

So,

$$c_1 \approx \frac{1.16}{0.773} \approx 1.50$$

Similarly, $c_2, c_3, \ldots$ can be computed. Typically, higher modes have smaller coefficients.

4. Transient Factor

$$F_n(\tau) = e^{-\tau r_n^2}$$

For $r_1 \approx 2.16$, at $\tau = 0.5$:

$$F_1(0.5) = e^{-0.5\times(2.16^2)} = \ e^{-2.34} \approx 0.096$$

5. Radial Function

$$J_0(r_n\xi)$$

At $\xi = 0, J_0(0) = 1$.
At $\xi = 0, J_0(r_1\ \cdot) = J_0(2.16) \approx -0.14$.

6. Series Solution for Temperature

$$\theta(\xi, \tau) = \sum_{n=1}^{N} c_n J_0(r_n\xi) e^{-\tau r_n^2}$$

Let us calculate **at center** ($\xi = 0$) and **first mode only**:

- $\xi = 0, J_0(0) = 1$.
- $\tau = 0.5$:

$$\theta(0{,}0.5) = c_1 F_1(0.5) = 1.50 \times 0.096 \approx 0.144$$

At $\tau = 0$, $\theta(0, 0) = c_1 \cdot 1 = 1.50$.
At **later time**, it decays exponentially.

Result

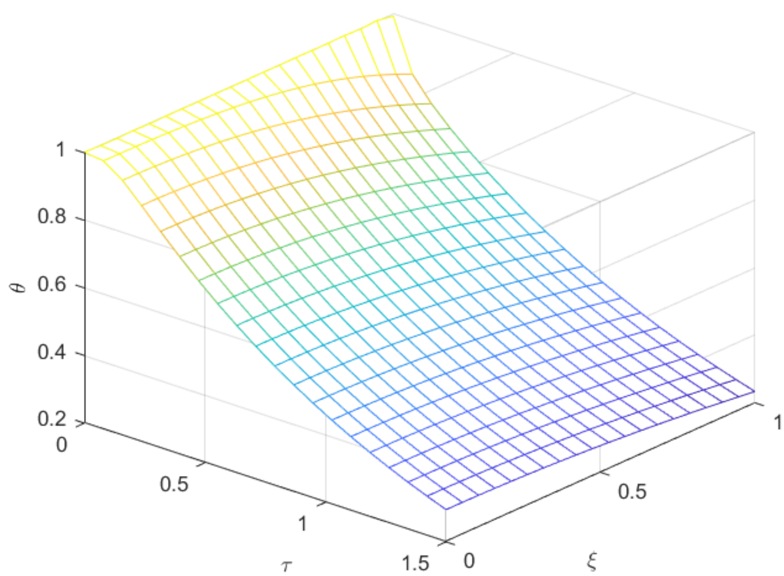

Figure 1 Temperature distribution in an infinite cylinder as a function of non-dimensional time τ and nondimensional radial position ξ for Bi = 0.5.